Homework 5 Solutions: Applications of Derivatives, Approximation, and Implicit Functions

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Homework Problems: Submit individually

6. Suppose a firm's cost function is
$$C(q) = 50 + 10q + q^2.$$
The firm currently produces $q_0 = 20$ units.

(a) The marginal cost function is
$$C'(q) = 10 + 2q.$$

(b) At $q = 20$,
$$C'(20) = 10 + 2(20) = 50.$$

(c) A linear approximation says
$$C(q) \approx C(20) + C'(20)(q - 20).$$
If output rises from 20 to 21, then
$$\Delta q = 1.$$
Therefore,
$$\Delta C \approx C'(20)\Delta q = 50(1) = 50.$$
So the estimated increase in cost is
$$50.$$

(d) The exact increase in cost is
$$C(21) - C(20).$$
First,
$$C(21) = 50 + 10(21) + 21^2 = 50 + 210 + 441 = 701.$$
Also,
$$C(20) = 50 + 10(20) + 20^2 = 50 + 200 + 400 = 650.$$
Therefore,
$$C(21) - C(20) = 701 - 650 = 51.$$

(e) Marginal cost gives an approximation to the change in total cost because $C'(q)$ measures the instantaneous rate of change of cost with respect to output. Around $q = 20$, the cost function is approximately linear, so
$$\Delta C \approx C'(20)\Delta q.$$
The approximation is not exact here because $C(q)$ is curved, not linear.

7. Suppose output is produced according to
$$Y = F(L) = 30L^{1/3}.$$
The current level of labor is $L_0 = 27$.

(a) The first derivative is

$$F'(L) = 30 \cdot \frac{1}{3} L^{-2/3} = 10L^{-2/3}.$$

The second derivative is

$$F''(L) = 10\left(-\frac{2}{3}\right) L^{-5/3} = -\frac{20}{3} L^{-5/3}.$$

(b) The linear approximation around $L_0 = 27$ is

$$F(L) \approx F(27) + F'(27)(L - 27).$$

First,

$$F(27) = 30(27)^{1/3} = 30(3) = 90.$$

Also,

$$F'(27) = 10(27)^{-2/3}.$$

Since

$$27^{1/3} = 3,$$

we have

$$27^{2/3} = 9.$$

Therefore,

$$F'(27) = \frac{10}{9}.$$

For $L = 28$,

$$L - 27 = 1.$$

Thus,

$$F(28) \approx 90 + \frac{10}{9}(1) = 90 + \frac{10}{9} = \frac{820}{9} \approx 91.1111.$$

(c) The quadratic approximation is

$$F(L) \approx F(27) + F'(27)(L - 27) + \frac{1}{2} F''(27)(L - 27)^2.$$

We already have

$$F(27) = 90$$

and

$$F'(27) = \frac{10}{9}.$$

Now compute $F''(27)$:

$$F''(27) = -\frac{20}{3}(27)^{-5/3}.$$

Since

$$27^{1/3} = 3,$$

we have

$$27^{5/3} = 3^5 = 243.$$

Therefore,

$$F''(27) = -\frac{20}{3} \cdot \frac{1}{243} = -\frac{20}{729}.$$

For $L = 28$, $L - 27 = 1$, so

$$F(28) \approx 90 + \frac{10}{9}(1) + \frac{1}{2}\left(-\frac{20}{729}\right)(1)^2.$$

Thus,

$$F(28) \approx 90 + \frac{10}{9} - \frac{10}{729}.$$

Numerically,

$$F(28) \approx 91.0974.$$

(d) The exact value is

$$F(28) = 30(28)^{1/3}.$$

Using a calculator,

$$28^{1/3} \approx 3.0366.$$

Therefore,

$$F(28) \approx 30(3.0366) = 91.098.$$

(e) The linear approximation gives approximately

$$91.1111,$$

while the quadratic approximation gives approximately

$$91.0974.$$

The exact value is approximately

$$91.098.$$

Therefore, the quadratic approximation is closer.

8. Suppose a firm faces price p and per-unit tax t. Its profit is

$$\pi(q,t) = pq - C(q) - tq,$$

where

$$C(q) = q^2.$$

(a) Substituting $C(q) = q^2$, profit is

$$\pi(q,t) = pq - q^2 - tq.$$

Differentiate with respect to q:

$$\frac{\partial \pi}{\partial q} = p - 2q - t.$$

The first-order condition is

$$p - 2q - t = 0.$$

(b) Solve the first-order condition:

$$p - 2q - t = 0.$$

Rearranging,

$$2q = p - t.$$

Therefore,

$$q^*(t) = \frac{p-t}{2}.$$

(c) Differentiate $q^*(t)$ with respect to t:

$$\frac{dq^*}{dt} = -\frac{1}{2}.$$

(d) If $p = 40$ and $t = 4$, then

$$q^*(4) = \frac{40-4}{2} = \frac{36}{2} = 18.$$

(e) If the tax rises from 4 to 5, then

$$\Delta t = 1.$$

Using linear approximation,

$$\Delta q^* \approx \frac{dq^*}{dt}\Delta t.$$

Since

$$\frac{dq^*}{dt} = -\frac{1}{2},$$

we get

$$\Delta q^* \approx -\frac{1}{2}(1) = -\frac{1}{2}.$$

Therefore,

$$q^*(5) \approx q^*(4) - \frac{1}{2} = 18 - \frac{1}{2} = 17.5.$$

In this case, the approximation is exact because $q^*(t)$ is linear in t.

(f) The derivative

$$\frac{dq^*}{dt} = -\frac{1}{2}$$

is negative. Economically, this means that when the per-unit tax increases, the firm's optimal output decreases. A one-unit increase in the tax lowers optimal output by one-half unit.

9. Suppose equilibrium in a market is described by the equation

$$D(p, a) = S(p),$$

where demand is

$$D(p, a) = a - 2p$$

and supply is

$$S(p) = 10 + p.$$

(a) The equilibrium condition is

$$a - 2p = 10 + p.$$

Move everything to one side:

$$a - 2p - 10 - p = 0.$$

Therefore,

$$F(p, a) = a - 10 - 3p.$$

The equilibrium condition is

$$F(p, a) = a - 10 - 3p = 0.$$

(b) Solve explicitly:

$$a - 10 - 3p = 0.$$

Then

$$3p = a - 10.$$

Therefore,

$$p^*(a) = \frac{a - 10}{3}.$$

(c) Differentiate the explicit solution:

$$\frac{dp^*}{da} = \frac{1}{3}.$$

(d) Using the implicit function theorem,

$$\frac{dp^*}{da} = -\frac{F_a}{F_p}.$$

From

$$F(p, a) = a - 10 - 3p,$$

we have

$$F_a = 1$$

and

$$F_p = -3.$$

Therefore,

$$\frac{dp^*}{da} = -\frac{1}{-3} = \frac{1}{3}.$$

This matches the derivative from the explicit solution.

(e) Economically,

$$\frac{dp^*}{da} = \frac{1}{3}$$

means that an increase in the demand-shift parameter a raises the equilibrium price. If a increases by 1 unit, the equilibrium price rises by $\frac{1}{3}$ unit.

10. **Slightly harder problem.** Suppose a consumer's indifference curve is given implicitly by

$$F(x, y) = x^{1/2} + y^{1/2} - 10 = 0.$$

(a) The partial derivatives are

$$F_x = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}},$$

and

$$F_y = \frac{1}{2}y^{-1/2} = \frac{1}{2\sqrt{y}}.$$

(b) By the implicit function theorem,

$$\frac{dy}{dx} = -\frac{F_x}{F_y}.$$

Therefore,

$$\frac{dy}{dx} = -\frac{\frac{1}{2\sqrt{x}}}{\frac{1}{2\sqrt{y}}}.$$

Simplifying,

$$\frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}}.$$

(c) At $(x, y) = (16, 36)$,

$$\frac{dy}{dx} = -\frac{\sqrt{36}}{\sqrt{16}} = -\frac{6}{4} = -\frac{3}{2}.$$

(d) If x increases from 16 to 17, then

$$\Delta x = 1.$$

Using linear approximation,

$$\Delta y \approx \frac{dy}{dx}\Delta x.$$

At $(16, 36)$,

$$\frac{dy}{dx} = -\frac{3}{2}.$$

Therefore,

$$\Delta y \approx -\frac{3}{2}(1) = -\frac{3}{2}.$$

So y must decrease by approximately

$$1.5.$$

The approximate new value of y is

$$y \approx 36 - 1.5 = 34.5.$$

(e) This makes economic sense. Along an indifference curve, if the consumer receives more of good x, then they must give up some of good y to remain at the same utility level. Therefore, y should fall when x rises.

(f) To find the exact value of y when $x = 17$, solve

$$17^{1/2} + y^{1/2} = 10.$$

Then

$$\sqrt{y} = 10 - \sqrt{17}.$$

Therefore,

$$y = (10 - \sqrt{17})^2.$$

Using a calculator,

$$\sqrt{17} \approx 4.1231.$$

Hence,

$$\sqrt{y} \approx 10 - 4.1231 = 5.8769.$$

Therefore,

$$y \approx (5.8769)^2 \approx 34.5379.$$

The exact change in y is approximately

$$34.5379 - 36 = -1.4621.$$

The linear approximation gave

$$-1.5.$$

The two answers are close, and the approximation correctly predicts that y decreases.